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Quantum convolutional neural network-based hybrid network for remote sensing image classification

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ABSTRACT

Remote sensing image classification is critical in geospatial applications such as land cover mapping and environmental monitoring. While traditional Convolutional Neural Network (CNN) have demonstrated strong performance in this domain, but show reduced effectiveness when dealing with complex spectral and spatial variations in remote sensing data, creating a clear research gap in achieving both high accuracy and computational efficiency. To address this gap, the present study proposes a novel Artificial Intelligence (AI)-driven Quantum Convolutional Neural Network-based Hybrid Network (QCHNet), which seamlessly integrates quantum computing principles with classical deep learning architectures to significantly enhance classification performance. The engineering application of this model is demonstrated in the context of land use and land cover (LULC) classification using Linear Imaging Self-Scanning Sensor-III (LISS-III) multispectral satellite imagery, implemented on the Matrix Laboratory (MATLAB) Quantum Computing platform. Experimental results show that QCHNet achieved an overall classification accuracy of 96.54 %, outperforming both classical CNN (90.70 %) and Quantum CNN (94.27 %). Performance metrics such as precision (96.4 %), recall (97.1 %), and F1-score (96.8 %) further validated its effectiveness, despite modest training (1.26 h) and testing (0.62 h) overheads. This work demonstrates how AI-driven quantum-classical hybrid models can be effectively applied to solve engineering problems in remote sensing, pushing the boundaries of classification accuracy in complex geospatial datasets.

1. Introduction

Remote sensing image classification plays a crucial role in Earth observation (EO), enabling the extraction of land use and land cover (LULC) information from vast amounts of satellite imagery (Li et al., 2020). However, with the rapid advancements in remote sensing technology, EO has entered the big data era, presenting significant challenges in processing and analyzing large-scale datasets (Pushpalatha et al., 2025). Traditional machine learning and deep learning approaches often struggle with the computational complexity and power required for training the model for accurate classification (Mahendra et al., 2024; Dou & Zeng, 2020). This limitation necessitates the exploration of alternative computing paradigms. One prospective option is quantum computing, which has the ability to tackle complicated problems substantially faster than classical computing. Applications for remote sensing could be completely transformed by its effective processing of high-dimensional data (Kharsa et al., 2023).

A new multidisciplinary field called Quantum Machine Learning (QML) blends machine learning methods with the ideas of quantum computing. Its potential has been investigated by researchers in a

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number of fields, such as quantum entanglement indication, data processing, segmentation, classification, and optimization (Ristè et al., 2015; Lu et al., 2021). As Noisy Intermediate-Scale Quantum (NISQ) technology has emerged, QML has gained significant attention for image classification applications. Image analysis is improved by quantum computing's capacity to speed up intricate calculations, while classification tasks requiring definitive results are well suited to QML algorithms' intrinsic probabilistic character (Gandhi et al., 2014).

Quantum annealers or parameterized quantum circuit (PQC) have been used in a number of QML-based image classification models. Among them, interest has grown in Quantum Neural Networks (QNNs), a branch of QML (Anand et al., 2022). However, their effectiveness compared to classical neural networks remains an open research question. Additionally, the application of QML for remote sensing image classification is still underexplored. Given the increasing demand for efficient remote sensing image analysis, further investigation into QML-based classification methods is necessary (Abohashima et al., 2020). QML holds immense potential to revolutionize computational tasks. However, the constraints of existing quantum devices make their practical implementation difficult. These devices only support a small number of qubits and are not completely fault-tolerant, which limits the scalability of QML algorithms (Perez-Salinas et al., 2220). Small-scale simulations are the only way to test the majority of proposed quantum algorithms (Nguyen et al., 2020). Despite these constraints, research in QML remains crucial, as it lays the foundation for future advancements when more powerful quantum devices become available, enabling broader real-world applications (Konar et al., 2022; Wilson et al., 2021; Abhishek et al., 2023).

Deep learning techniques, especially convolutional neural network (CNN), have greatly helped remote sensing image classification because of their remarkable feature extraction capabilities (Zhu et al., 2017). When processing data from remote sensing, CNN have shown exceptional performance and adaptability. However, limited research has explored the quantum variant of CNNs, which could harness quantum computing's potential for enhanced classification accuracy (Shi et al., 2018, 2020). This study investigates a quantum CNN model designed to efficiently extract critical features while addressing the constraints of available qubits in quantum devices. By optimizing qubit usage, the proposed model aims to advance quantum-assisted remote sensing image classification, thereby contributing to both quantum computing and geospatial data analysis domains.

To obtain precise LULC mapping in remote sensing, researchers have explored diverse methodologies and data sources. Bouslihim et al. (2022) assessed the effectiveness of Landsat-9 imagery for producing LULC maps. Pan et al. (2020) introduced a convolutional neural network-based approach that leverages multi-spectral LiDAR data as a novel input for LULC classification tasks. Meanwhile, Dou et al. (2021) proposed a technique called feature relations map learning (FRML), which enhances the distinction between various elements within an image to improve classification performance. Similarly, Chen et al. (2023) incorporated attention mechanisms across channels within the feature fusion component of a U-Net model to analyze Sentinel-2A imagery effectively.

Furthermore, several studies Yao et al. (2023) and He et al. (2022) have moved beyond single-date imagery by utilizing time-series Earth Observation data, aiming to enhance classification reliability. Beyond relying on individual classifiers, research such as that by Dou et al. (2021) has explored ensemble methods, combining multiple classifiers to harness their complementary strengths for improved LULC prediction. However, the incorporation of advanced mechanisms in deep learning models, while improving accuracy, significantly raises the computational demand required for training. Concurrently, the progression of remote sensing technologies has ushered the field into the big data era, leading to an exponential increase in EO datasets. This surge poses scalability challenges for conventional computing systems (Cheng et al., 2020).

One interesting method for improving deep learning model performance is quantum computing. While prior studies have demonstrated the potential of quantum computing and hybrid quantum-classical neural networks (Fan et al., 2023; Hur et al., 2022; Sebastianelli et al., 2022; Li et al., 2020), these works primarily evaluated models on benchmark datasets such as MNIST, CIFAR, or small-scale medical images. In contrast, our study introduces Quantum-CNN-based hybrid network model (QCHNet), for remote sensing image classification using Linear Imaging Self-Scanning Sensor-III (LISS-III) satellite data. Unlike existing QCNN approaches, QCHNet integrates optimized quantum convolutional layers with a lightweight CNN backbone, enabling efficient extraction of spatial-spectral features from high dimensional geospatial imagery. Furthermore, we present a comprehensive comparative evaluation with multiple state-of-the-art quantum-classical models, demonstrating consistent improvements in accuracy, kappa values, and computational efficiency. This makes QCHNet one of the systematically validated quantum-classical frameworks for satellite-based land cover classification, thereby contributing a meaningful advancement to both the fields of quantum machine learning and remote sensing.

We conducted experiments using the Matrix Laboratory (MATLAB) Quantum Computing platform and used LISS-III multispectral satellite data covering Hunsur taluk, Mysuru district, Karnataka, India, to assess the performance of the proposed model. According to the experimental results, the QCHNet model performs better and has more generalizability than its classical version. To learn more about the characteristics and resilience of the model, we also methodically investigated the effects of different quantum gates, measurement methods, and model topologies. These results confirm the effectiveness of the model and highlight how quantum computing can be used to improve LULC classification results. The novelty of our work lies primarily in the integration of Flexible Representation of Quantum Images (FRQI) for efficient amplitude encoding of grayscale remote sensing images. FRQI allows a compact representation of both pixel intensity and position information within quantum states, which reduces the required number of qubits while preserving essential spatial information. This encoding is particularly beneficial for high-dimensional remote sensing data. The results demonstrate that FRQI-based quantum convolution enhances classification accuracy by 2–4 % over classical CNNs, particularly in cases where spectral-spatial correlations are complex. This provides concrete evidence that the quantum module contributes meaningfully to feature extraction. This study's primary contributions are as follows:

1. We introduce QCHNet, a hybrid quantum–classical model for multispectral image classification that leverages quantum circuits with deep learning to enhance feature extraction and improve accuracy.
2. We design a quantum convolution layer that reduces qubit requirements, ensuring model simplicity without compromising classification performance.
3. We evaluate QCHNet on LISS-III satellite imagery of Hunsur taluk, where it outperforms CNN and QCNN in terms of accuracy, precision, recall, and F1-score.

2. Background

Quantum computing exploits the principles of superposition and entanglement to perform computations beyond the reach of classical systems. Unlike classical bits, which represent either 0 or 1, a quantum bit (qubit) exists in a superposition of both states. Equation (1) represents a qubit state mathematically:

$$|\Psi\rangle = \gamma\,|0\rangle + \delta\,|1\rangle;\ |\gamma|^2 + |\delta|^2 = 1.\ \gamma, \delta \in C \tag{1}$$

Where complex coefficients γ and δ represent probability amplitudes. Upon measurement, the qubit collapses to either $|0\rangle$ or $|1\rangle$, allowing quantum systems to explore multiple computational paths in parallel. Entanglement further enriches this capability, creating correlations

between qubits that cannot be factorized into independent states, thereby enabling powerful joint computations.

The widely adopted quantum circuit model comprises three stages: state initialization, gate operations, and measurement. Initialization transforms classical data into quantum states using techniques such as computational basis encoding or amplitude encoding, the latter leveraging normalized amplitudes to represent data efficiently (Su et al., 2020).

In the second stage, quantum gates manipulate qubit states through unitary operations that preserve reversibility. Gates such as Hadamard, Pauli-X, RY, and parameterized gates (U3, CU3) enable superposition, entanglement, and flexible rotations. Together, these form the basis of parameterized quantum circuits (PQCs), where rotation parameters act as trainable variables analogous to neural network weights.

Finally, measurement extracts classical outcomes from quantum states, typically as expectation values. These serve as features in QML models. Hybrid quantum–classical architectures employ PQCs alongside classical optimizers, allowing effective learning while mitigating current hardware constraints (Ovalle-Magallanes et al., 2022; Huang et al., 2022; Zheng et al., 2022).

In this work, PQCs constitute the computational core of the quantum convolutional layer in our proposed QCHNet model. To bridge classical multispectral image data with quantum computation, the FRQI is adopted, as it enables efficient amplitude encoding of pixel intensities into quantum states. Once encoded, the data is processed through PQCs, where superposition allows parallel representation of multiple pixel states and entanglement captures complex spatial–spectral dependencies. By integrating PQCs with classical CNN layers under a hybrid training scheme, QCHNet achieves enriched feature representations and improved classification accuracy, demonstrating the promise of quantum-enhanced remote sensing within current NISQ-era limitations.

3. Related work

Recent advancements have sparked significant interest in incorporating quantum computing into image classification. Numerous studies have examined the possibilities of quantum deep learning and machine learning methods for image classification, with encouraging outcomes. Furthermore, recent studies have looked into how quantum computing may be used in the EO field, emphasizing how it could improve data processing and analysis. This section offers an informative summary of these contributions, first looking at methods that use quantum-based learning models to classify images before going over more recent advancements in the use of quantum computing in the EO field.

3.1. Quantum-based learning models

Within the QML community, quantum image classification has attracted a lot of attention, resulting in a number of creative methods. The quantum support vector machine (QSVM), which was first presented by Rebentrost et al. (2014) and Havlíček et al. (2019), is one of the early efforts in this field. Their study showed how classical support vector machines may be improved by quantum computing, allowing for more effective classification tasks. Other quantum-enhanced machine learning methods have been investigated in addition to QSVM. A quantum model that uses principal component analysis (PCA) for image classification was presented by Ostaszewski et al. (2015). They improved classification performance by efficiently reducing the dimensionality of image data through the use of quantum methods.

Liu et al. (2023) presented a quantum-enhanced machine learning, specifically a support vector machine-quantum annealing solver (SVM-QA), which outperformed classical algorithms in satellite image classification for 10 out of 16 datasets. Lin et al. (2024) implemented a quantum adversarial learning (QAL) model for hyperspectral image processing in remote sensing, addressing challenges such as limited qubit resources and quantum collapse effects. Shaik et al. (2022) introduced a quantum-based pseudo-labeling for hyperspectral imagery to prepare datasets for machine learning algorithms, achieving 86 % accuracy in pine tree classification using a hybrid XGBoost and decision tree technique.

Miller et al. (2024) present a novel quantum image fusion technique that enhances the spectral details of C-band SAR and optical images, improving object classification compared to using raw SAR data alone. To further enhance the classification results, the research also explains how to employ a modified deep neural network, namely the VGG16 convolutional neural network. Delilbasic et al. (2021) presented a quantum formulation of the SVM. The implementation of these quantum SVM algorithms using existing quantum technologies was discussed. The performance of the quantum SVM algorithms was evaluated for remote sensing image classification tasks.

The combination of quantum computing and the K-nearest neighbor (KNN) method is another significant advancement in this area. In order to create an effective counterpart of the conventional KNN algorithm, Ruan et al. (2017) created a quantum KNN (QKNN) model in which they calculated Hamming distances in a quantum framework. By adding the quantum minimum search algorithm to their QKNN model, Dang et al. (2018) improved this strategy even further. Their approach demonstrated the promise of quantum computing in optimizing conventional machine learning algorithms by increasing the speed of similarity search processing without sacrificing classification accuracy.

3.2. Quantum neural networks

Utilizing the potential of quantum computing to improve on classical CNNs, Quantum Neural Networks (QNNs) have been extensively studied for the classification of images and recognition. CNNs, which are renowned for their capacity to sequentially filter images in order to extract high-level features, have shown encouraging results in a range of classification applications. In order to create QCNNs, researchers have made great progress in combining CNNs with quantum computers. A QCNN design with successive convolution and pooling layers was proposed by Cong et al. (2019). Their approach showed promise for phase classification in quantum physical systems by combining quantum error correction and multiscale entanglement renormalization ansatz. QCNNs for identifying topological quantum phases were experimentally validated by Herrmann et al. (2022), further demonstrating the viability of quantum-enhanced classification models.

QCNNs have been explored in a number of studies for the classification of images. Before using QCNNs for classification, Chen et al. (2020) preprocessed images using a fractal scaling dimension reduction approach. A quantum state preparation method has been proposed by Lu et al., (221) in order to encode input images and use QCNNs for classification. The success of these quantum methods was confirmed by the experimental findings in both investigations. Using the MNIST dataset, Kerenidis et al. (2019) created a quantum algorithm that might achieve accuracy levels comparable to those of traditional CNNs. For digit recognition tasks, Wei et al. (2022) presented a QCNN architecture with fewer parameters and showed classification performance comparable to that of classical CNNs. Using classical preprocessing methods like PCA and autoencoders to extract initial global features prior to quantum processing, Hur et al. (2022) introduced a fully parameterized QCNN intended for classical data classification. In order to improve local feature extraction, Henderson et al. (2020) presented quanvolutional neural networks with a quantum transformation layer (quanvolution layer) based on random quantum circuits. On the MNIST dataset, their tests demonstrated that quanvolutional networks might outperform traditional CNNs in terms of accuracy and efficiency. However, the consistency of results was limited by an unrepeatable process imposed by the unpredictability of quantum circuits.

Recent advancements have led to hybrid quantum-classical CNN architectures. In order to improve feature extraction and classification

performance across a range of tasks, Matic et al. (2022) and Chen et al. (2022) independently presented hybrid CNN models that use parameterized quantum circuits (PQCs) for convolutional kernels rather than random quantum circuits. Similarly, in order to efficiently convert image features without the need for extra trainable parameters, Riaz et al. (2023) used strongly entangled quantum circuits as kernels. Hrithik et al. (2024) presented Quantum-CNN-based hybrid models that show promise for remote sensing image classification, outperforming classical CNN, but struggle with complex feature segmentation compared to U-Net. The FQCNN quantum model outperformed the classical CNN model by 8 % in remote sensing classification accuracy. Quantum models had quicker convergence but struggled with complex feature segmentation compared to the U-Net classical model. Quantum convolutions show potential to enhance convergence but need to address challenges like noise from quantum channels to improve accuracy.

Despite these developments, binary classification tasks remain the main focus of many current research. In order to overcome this restriction, Zeng et al. (2022) proposed a hybrid QNN that uses a conventional dense layer for multicategory classification and a ladder-like PQC for input image encoding. However, their approach had scalability issues because it did not account for spatial information in feature extraction and needed more qubits as the image size rose. The promise of quantum computing to advance the classification of images is demonstrated by current studies in QNNs and QCNNs. However, there is still ongoing research into issues like scalability, encoding methods, and computer resource requirements.

3.3. QML and QNNs in the domain of remote sensing

In the field of remote sensing, QML and QNNs have drawn a lot of interest, especially for image classification tasks (Nielsen and Chuang, 2010). In order to improve classification performance and computational efficiency, a number of studies have investigated the integration of quantum computing with traditional techniques (Boyda et al., 2017; Cavallaro et al., 2020). Using quantum annealers for classification tasks is one main strategy. Researchers have shown the potential benefits of quantum computation in managing complicated datasets by examining the application of quantum annealing approaches to the classification of remote sensing images (Otgonbaatar and Datcu, 2021). Quantum circuits for the analysis of remote sensing data have been the subject of another research stream. However, the majority of current approaches use quantum circuits mainly for final classification and classical algorithms for feature extraction. Gawron and Lewiński (2020) proposed a neural network based on quantum circuits for classifying multispectral images in land cover mapping. Prior to using quantum processing, their strategy included PCA as a traditional feature reduction technique. In a similar vein, Zaidenberg et al. (2021) presented a hybrid model that used a quantum circuit for the final classification of EO data and a CNN for feature extraction. For LULC classification, Sebastianelli et al. (2022) developed a circuit-based hybrid QNN model in which high-level characteristics were extracted using a classical CNN, modeled LeNet-5, and then a quantum layer was included for classification.

The work of Abdel-Khalek et al. (2023) is another example of further developments in hybrid models. They used a QNN for classification after extracting features from high-resolution images using an Inception ResNet architecture. Their experimental findings proved that the proposed quantum-enhanced model worked effectively. Furthermore, a hybrid method for evaluating polarimetric synthetic aperture radar images was introduced by Otgonbaatar and Datcu (2022). Their model used a quantum circuit to analyze the image's stokes parameters and extract characteristics to do pixel-by-pixel classification. The increasing interest in QML and QNNs for remote sensing applications is demonstrated by these works. Despite the potential benefits of quantum computing, the majority of existing methods rely on hybrid models that combine quantum classification with classical feature extraction.

Fan et al. (2023) introduced the QC-CNNs, which demonstrated superior results compared to classical CNNs on various Earth observation datasets, including EuroSAT, Overhead-MNIST, and So2Sat LCZ42. The quantum components, particularly those exploiting entanglement, contribute to improved classification scores. Additionally, QC-CNNs can accelerate convolutional operations and reduce computational resources through techniques like amplitude encoding. Transfer learning approaches incorporating pre-trained CNNs and quantum circuits have shown promise in scenarios with limited labeled data (Kim et al., 2023). These advancements highlight the potential of quantum computing in addressing the computational challenges posed by big data in Earth observation applications.

Zollner et al. (2024) identify autoencoder techniques that are appropriate for pre-processing. Remote sensing images were classified using hybrid classic-quantum systems that combined classical and quantum hardware. The study assessed several circuit topologies and pre-processing methods in the hybrid systems. It emerged that autoencoder techniques were especially well-suited for pre-processing when applying quantum machine learning to remote sensing applications. For effective multi-spectral remote sensing image classification, Fan et al. (2024) explored quantum neural network classifiers. The study examined the classification of multispectral Earth observation data using quantum circuit-based neural networks to determine land cover. The article offers a proof-of-concept illustration of their method for classifying multispectral images using quantum neural networks.

The studied research works in the literature show promising improvements in accuracy and efficiency by integrating quantum computing with image classification. Quantum models like QSVM, QKNN, and QCNNs have enhanced classification in both general and remote sensing applications. Hybrid quantum-classical models effectively combine classical CNNs with quantum circuits for hyperspectral and multispectral image analysis. However, challenges such as scalability, limited qubit resources, and inconsistent results due to quantum randomness remain. These limitations highlight the need for more robust and scalable quantum solutions like the proposed QCHNet.

4. Materials and methods

4.1. Objectives

1. To develop a hybrid QCHNet model that integrates quantum feature extraction with classical classification for improved remote sensing image classification.
2. To evaluate the performance of the proposed QCHNet model using LISS-III satellite imagery and compare it with classical CNN and QCNN models in terms of accuracy, precision, recall, and F1-score.
3. To conduct a comparative analysis with existing state-of-the-art Quantum-CNN models and demonstrate the performance of QCHNet in real-world Earth observation applications.

4.2. Study area

The study area identified for this work is Hunsur taluk, located in Mysuru district, Karnataka, covering approximately 1991 square kilometers and featuring a diverse landscape comprising agricultural fields, forests, water bodies, and urban settlements, making it a suitable study area for remote sensing image classification. Geographically positioned between 12.30°N and 12.55°N latitude and 76.15°E to 76.50°E longitude, the region experiences a semi-arid climate with an average annual rainfall of 800-1000 mm, primarily during the monsoon season. Agriculture dominates the local economy, with major crops including paddy, ragi, maize, sugarcane, and tobacco, supported by the Cauvery River and its tributaries. Additionally, the taluk houses significant forest reserves, including parts of Nagarahole National Park, contributing to its ecological richness. The rapid expansion of urban settlements and land use changes poses challenges for accurate classification using remote

sensing techniques. The study area location is shown in Fig. 1.

4.3. Dataset

This research utilizes a multi-temporal LISS-III image of the year 2020, obtained from the National Remote Sensing Centre (NRSC), ISRO, covering the study area. The LISS-III dataset from the IRS satellite provides multispectral imagery crucial for LULC classification in Hunsur Taluk, Mysuru district, Karnataka. Capturing data in green, red, near-infrared (NIR), and shortwave infrared (SWIR) bands at a 23.5 m spatial resolution, LISS-III enables accurate analysis of agricultural land, forests, water bodies, and urban expansion. Preprocessing steps, including radiometric and geometric corrections, atmospheric correction, and cloud removal, ensure high-quality data for analysis. By integrating LISS-III imagery with the QCHNet model, this research enhances LULC classification accuracy, leveraging the sensor's spectral capabilities to differentiate land cover types effectively. The study aims to monitor spatial-temporal land cover changes and support sustainable land management, demonstrating the potential of remote sensing and quantum-inspired deep learning in geospatial analysis.

4.4. Methodology

The methodology followed in the research paper involves a structured pipeline for processing multispectral data and training the proposed quantum-classical hybrid model, as shown in Fig. 2. The process begins with the acquisition of multispectral data, which serves as the primary input for classification. These remote sensing images consist of multiple spectral bands that provide detailed information about different land cover types. The first step in the methodology involves data preprocessing and band composition, which includes radiometric and geometric corrections, noise removal, and spectral band selection to enhance relevant features. This step ensures that the input data is optimized for classification and helps improve the model's performance.

Following preprocessing, the dataset is split into three subsets in a 60:20:20 ratio, where 60 % of the data is used for training, 20 % for validation, and 20 % for testing. The training sample is used to train the proposed QCHNet model, while the validation samples help fine-tune the model's hyperparameters and prevent overfitting. The test samples are reserved for the final evaluation to assess the model's generalization ability. This structured data splitting ensures that the model learns effectively while being evaluated on unseen data for unbiased performance assessment.

Once the dataset is prepared, the QCHNet model is implemented using the training and validation samples. The model, which integrates quantum convolutional layers with classical deep learning, extracts features from multispectral images using quantum operations and then processes them through classical dense layers for classification. The hybrid approach aims to leverage quantum parallelism for efficient feature extraction while retaining the learning capabilities of classical neural networks. The classification module assigns each pixel or image to a specific land cover class, such as urban areas, vegetation, water bodies, or agricultural fields.

After classification, the model evaluation phase is conducted using test samples. Various performance metrics are used to assess the effectiveness of QCHNet, including overall accuracy, Kappa value, precision, recall, and F1 score. These metrics provide insights into the model's classification accuracy, reliability, and ability to distinguish between different land cover types. The Kappa value measures agreement between predicted and actual classifications, while precision and recall evaluate the model's ability to correctly identify different classes. The F1 score, a harmonic mean of precision and recall, ensures a balanced assessment of model performance.

4.5. Proposed QCHNet model

The proposed QCHNet model is a hybrid Quantum-CNN-based network designed for remote sensing image classification, combining the power of quantum computing with classical deep learning, as shown in Fig. 3. Traditional deep learning models, such as CNNs, have been widely used for remote sensing applications but often require high computational resources for processing large-scale satellite imagery. By integrating quantum computing, QCHNet aims to leverage the advantages of quantum parallelism and reduce computational complexity while maintaining high accuracy in image classification tasks. Quantum convolution exploits the principles of superposition and entanglement to

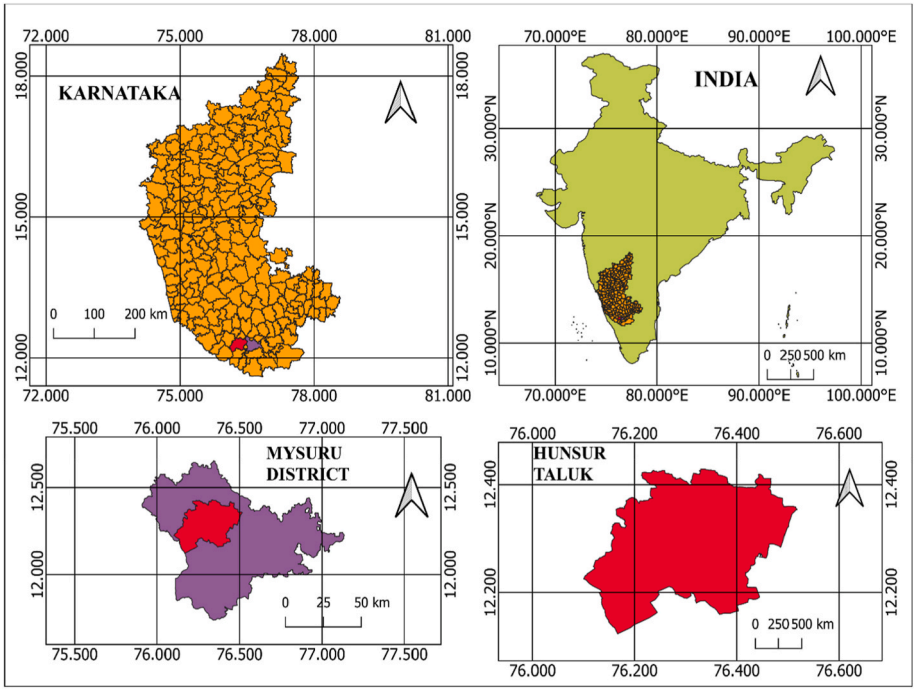


Fig. 1. Study area.

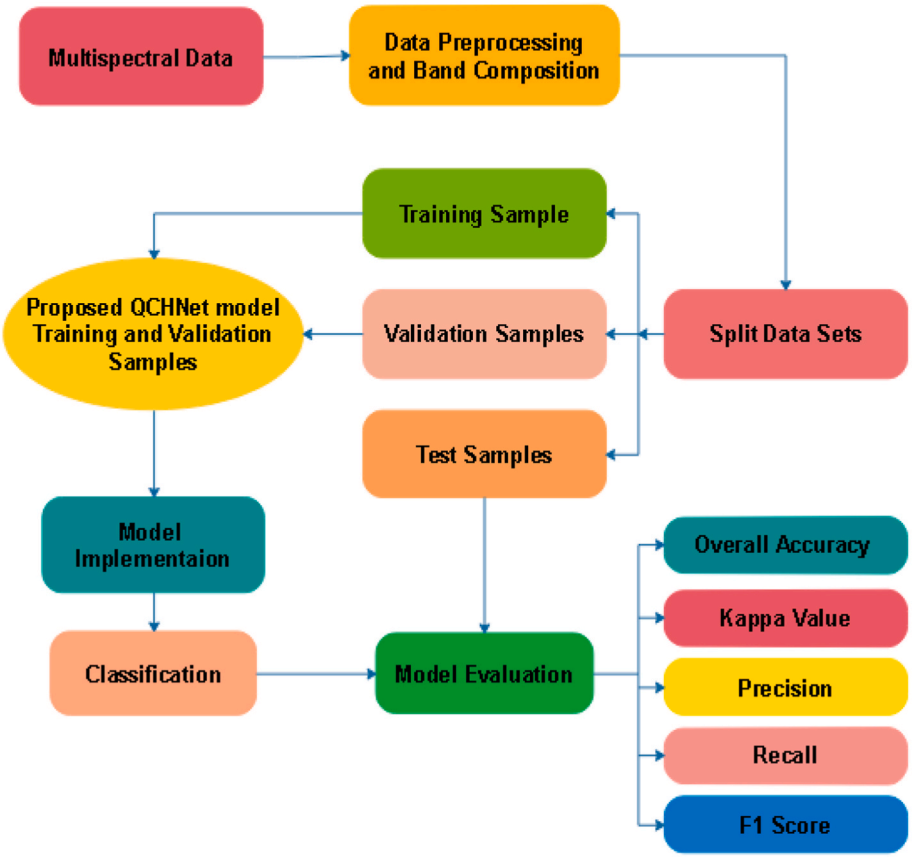


Fig. 2. Methodology.

represent and process multiple pixel states simultaneously. Superposition allows the parallel encoding of spectral–spatial features, while entanglement captures correlations between spatially distant pixels, enabling the detection of textures, boundaries, and edges that are often difficult for classical convolutions to capture. This richer feature representation explains the improved classification accuracy of the hybrid model, even under limited qubit resources.

In classical deep learning, convolution operations are applied over local regions of an image to extract spatial features. However, directly executing such classical convolutions on quantum superposition states is not feasible due to the linearity of quantum mechanics and the absence of sliding-window style operations in quantum circuits. To address this challenge, we reformulate convolution in the quantum domain using parameterized quantum circuits. Specifically, quantum filters are implemented through parameterized rotation gates (e.g., RY, RZ) and controlled entangling operations, which act simultaneously across the superposition of pixel states. This design leverages quantum parallelism, enabling feature extraction across the entire image representation without the need for explicit classical convolution. By mapping convolutional kernels to parameterized gates, the model preserves the feature learning capability of classical CNNs while remaining compatible with the constraints of quantum computation.

The architecture of QCHNet begins with an input image (NxNxM), where N represents the spatial resolution, and M refers to the number of spectral channels. The first stage is the Encoding Layer, which transforms the classical image into a quantum image representation using encoding techniques like FRQI or Normalized Extended Quantum Representation (NEQR). These encoding methods store pixel intensities as quantum states, enabling efficient data representation and manipulation in a quantum computing framework. This transformation allows QCHNet to take advantage of quantum parallelism, where multiple computations can be performed simultaneously.

Once the image is encoded, it is passed through the Quantum Convolution Layers, which perform quantum convolution operations to extract essential features from the image. These layers use quantum circuits instead of traditional convolutional filters, reducing the number of required parameters and computations compared to classical CNNs. Quantum convolution can detect spatial patterns, textures, and edges in the image while benefiting from the exponential speedup offered by quantum algorithms. This step is crucial in remote sensing applications where high-dimensional data requires effective feature extraction.

After feature extraction, the measurement layer is introduced to convert the quantum-extracted features into classical values by measuring quantum states. Since quantum information cannot be directly interpreted in a classical computing environment, this layer plays a critical role in bridging the quantum and classical components of the model. The extracted feature maps are then fed into a dense layer, which is a fully connected classical neural network that processes the features and classifies the image into predefined land cover categories such as urban areas, forests, water bodies, and agricultural lands.

Finally, the model produces the output classification result, determining the category of the input remote sensing image. By combining quantum computing for feature extraction with classical deep learning for classification, QCHNet offers an innovative and computationally efficient approach for processing large-scale remote sensing data. The hybrid nature of QCHNet allows it to outperform traditional CNNs in terms of speed and scalability while maintaining or even improving classification accuracy. This makes QCHNet a promising model for future applications in Earth observation, land cover mapping, and environmental monitoring, where efficient and accurate classification of remote sensing imagery is essential.

The subsequent sections provide a detailed explanation of each layer's function and the overall data flow. In the QCHNet model, qubits are categorized based on their encoded information to facilitate

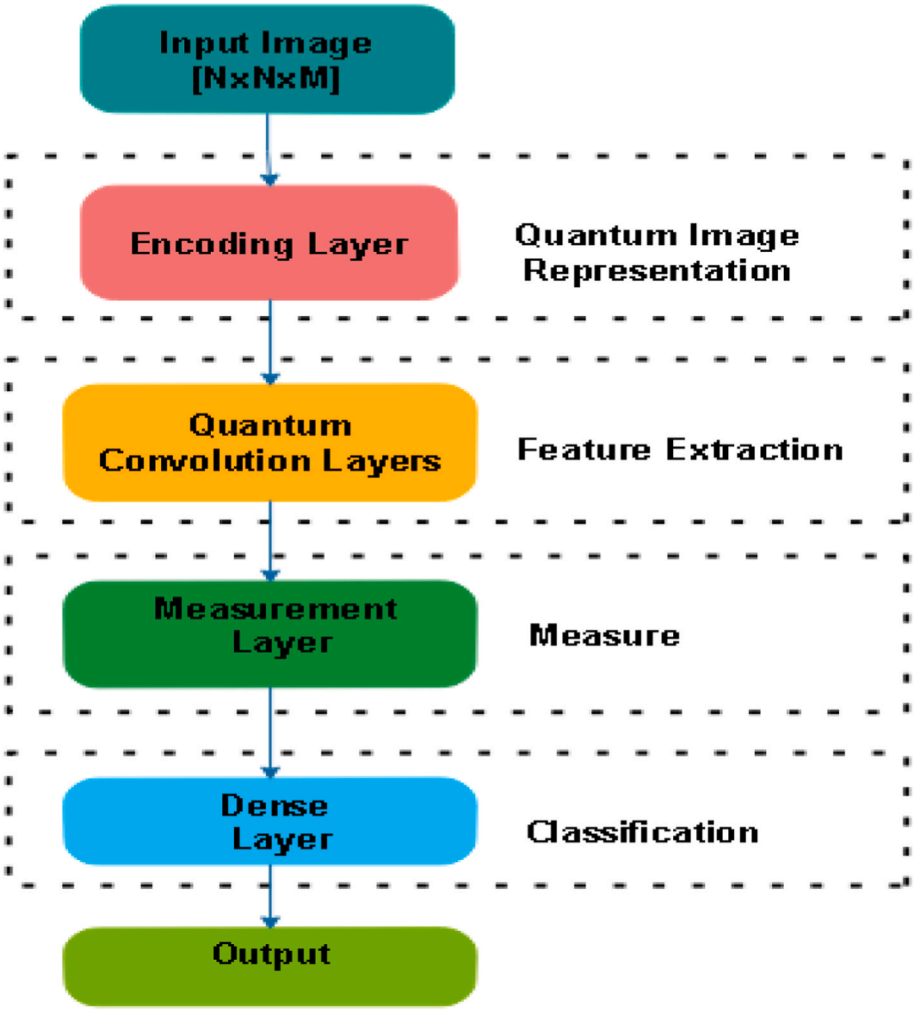


Fig. 3. Proposed QCHNet model.

structured quantum computation. The different groups of qubits serve distinct roles: Sq qubits encode the spatial information of input images, Cq qubits store color information, Kq qubits represent quantum convolution kernels, and Fq qubits represent the quantum feature maps. These qubits interact within the quantum layers to extract and process relevant features before being measured and passed to the classical dense layer for classification.

4.5.1. Quantum image representation using the Encoding Layer

The Encoding layer is a fundamental component of the quantum model, responsible for transforming classical image data into quantum states for further processing. In this work, the FRQI is adopted, as it enables efficient amplitude encoding of grayscale images. For an image of size $2^n \times 2^n$, each pixel is uniquely identified by its coordinate pair (x, y), which is encoded into a quantum basis state $|Q_{x,y}\rangle$. By applying Hadamard gates, the qubits are placed into a superposition of all possible coordinates, as represented in Equation (2):

$$\frac{1}{2^n}\sum_{x=0}^{2^n-1}\sum_{y=0}^{2^n-1}|Q_{x,y}\rangle \tag{2}$$

Where $|Q_{x,y}\rangle$ represents the binary string form coordinates (x, y) of a pixel.

Once the coordinates are encoded, grayscale intensity values are mapped to amplitudes using a single qubit (Cq) per pixel. Each pixel value in the range [0,255] is normalized to an angle $\theta_{x,y} \in [0, \pi/2]$ as respresented in Equation (3).

$$\theta_{x,y} = \frac{\pi}{2} \cdot \frac{I(x,y)}{255} \tag{3}$$

This angle is then applied to a single color qubit using the RY gate as represented in Equation (4):

$$|C_{x,y}\rangle = \cos\theta_{x,y}|0\rangle_C + \sin\theta_{x,y}|1\rangle_C \tag{4}$$

To bind pixel coordinates with their corresponding intensities, entanglement is established between the registers $|Q_{x,y}\rangle$ and $|C_{x,y}\rangle$ using multi-controlled RY(θ) gates. This results in the final quantum state in Equation (5), where all pixel information is stored in superposition:

$$(\psi_i) = \frac{1}{2^n}\sum_{x=0}^{2^n-1}\sum_{y=0}^{2^n-1}|Q_{X,Y}\rangle\,|C_{X,Y}\rangle \tag{5}$$

This formulation ensures appropriate normalization through amplitude encoding and enables efficient quantum image representation. Compared to computational basis encoding, which requires 2n+8 qubits, FRQI provides a compact representation with only 2n+1 qubits. Its implementation complexity, however, increases rapidly with image size. For example, a 32×32 image requires 11 qubits and preparation of 1024 pixel amplitudes through numerous controlled rotations, resulting in exponential gate complexity. Such deep circuits are highly sensitive to noise and decoherence, making large-scale FRQI computationally demanding on current quantum hardware.

To balance feasibility and resource constraints, the input 128 × 128 image is downsampled to 32 × 32 to balance computational feasibility with quantum resource constraints. The same encoding principle applies to the 32 × 32 representation, requiring 2n+1 = 11 qubits for spatial and intensity encoding. However, increasing the encoded resolution inherently increases gate complexity and circuit depth due to the large number of controlled rotations. To improve efficiency, a patch-wise encoding strategy is employed, which minimizes quantum resource usage while preserving the fidelity of encoded features. This approach provides an effective trade-off between scalability and practical feasibility on current NISQ devices.

Fig. 4 shows an example of a 2 × 2 image implementation using the FRQI circuit. In this circuit, dot markers indicate controlled states, where white dots represent control on $|0\rangle$ and black dots represent control on $|1\rangle$. This controlled operation ensures that pixel location and color information are correctly mapped into the quantum state. By applying controlled rotations such as Multi-Controlled RY (θ), the circuit entangles pixel coordinates with their corresponding color values, allowing the entire image to be stored in a superposition.

Fig. 4 illustrates that the spatial qubits *Sq(Y)* encode the pixel coordinates, whereas the color qubit *(Cq)* stores the pixel intensity through controlled rotational operations. Initially, all qubits are initialized to the ground state $|0\rangle$, and hadamard gates are applied to the spatial qubits to create an equal superposition of all pixel positions, as respresented in Equation (6):

$$|\psi_0\rangle = \frac{1}{2}\sum_{i=0}^{3} |i\rangle_p \otimes |0\rangle_c \tag{6}$$

Subsequently, each pixel intensity is encoded by a controlled rotation $R_Y(\theta i)$ on the color qubit, where θ_i corresponds to the pixel's grayscale value. This produces the final quantum image representation shown in Equation (7):

$$|\psi\rangle = \frac{1}{2}\sum_{i=0}^{3}\left(\cos\frac{\Theta_i}{2}\ |0\rangle_C + \sin\frac{\Theta_i}{2}\ |1\rangle_C\right) \otimes |i\rangle_p \tag{7}$$

4.5.2. Feature extraction using quantum convolution layer

In the proposed QCHNet model, the quantum convolution layer serves as a fundamental component for performing convolutional operations, enabling efficient feature extraction and the generation of corresponding feature maps. This layer operates analogously to classical convolutional layers in CNNs but leverages quantum principles to enhance computational efficiency.

The quantum convolutional layer in QCHNet employs PQCs designed with a 2 × 2 kernel mapping to four qubits. Each filter consists of four U3 rotation gates per qubit, yielding up to 12 trainable angles per layer. Entanglement is introduced through controlled-NOT (CNOT) operations, with three entangling gates per 2 × 2 patch. For an architecture with F filters, the total gate count per layer is approximately 4 F single-qubit rotations and 3 F two-qubit entangling gates, resulting in a circuit depth ≤4 across all layers.

The inclusion of multi-qubit entangling gates significantly enhances expressivity, enabling non-trivial feature extraction that is not efficiently replicable by shallow classical counterparts. While exact classical simulation of small circuits is feasible, the scaling of entangling operations with depth and filter count rapidly increases the Hilbert space dimensionality (2^n for n qubits). This exponential state space, combined with variational optimization of PQCs, places the proposed design outside the regime of efficiently classically simulable circuits, aligning with current literature on quantum advantage for variational quantum models.

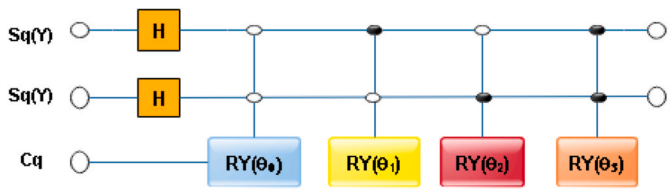


Fig. 4. Encoding layer.

The quantum convolution layer employs kernels of a predefined size to traverse the input image, applying convolution operations to extract spatial features. In this work, a QCHNet model utilizing a 2 × 2 kernel with a convolutional stride of 2 is demonstrated. The kernel size and stride are crucial hyperparameters that influence the dimension reduction of the feature maps and should be adjusted based on the input image size, similar to classical CNN architectures.

The feature extraction process involves systematically positioning the kernel over different regions of the input image. At each location, two main operations occur: first, an elementwise product is performed between the pixel values in the selected image patch and the corresponding kernel weights. Second, the resulting values from this multiplication are summed to produce the feature map output. In the quantum implementation of this process, a quantum register Rq is introduced, and controlled rotation gates are applied to encode the elementwise product operation. This approach enables efficient parallel computation, a key advantage of quantum computing over classical methods. Mathematically, the quantum state after performing the elementwise product can be expressed as shown in Equation (8):

$$(\psi_{fi}) = \frac{1}{2^n}\sum_{x=0}^{2^n-1}\sum_{y=0}^{2^n-1} |Q_{X,Y}\rangle\ \ |C_{X,Y}\rangle\ |R_{X,Y}\rangle \tag{8}$$

Where $(R_{x\,y})$ represents the weight of the kernel for the corresponding pixel. This quantum-based approach facilitates efficient feature extraction by leveraging quantum parallelism, thereby enhancing the representation of spatial features within the QCHNet model.

In our quantum convolution layer, we employ controlled U3 gates that rotate a qubit using three Euler angles, optimized during training. These gates function as trainable parameters, representing kernel weights. With a kernel size of 2 × 2, four U3 gates are required per kernel. To differentiate the spatial positions of these weights, two control qubits are introduced. Fig. 5 illustrates a quantum circuit representation for the QCHNet model. It consists of a single quantum convolution layer with one filter and two kernels per filter designed to process an input image of size 4 × 4. As shown in Fig. 5, each feature map is composed of features extracted from two spectral bands, ensuring comprehensive feature representation. Consequently, after the convolution process, two distinct feature maps of size 2 × 2 are generated, capturing essential spatial and spectral patterns. This quantum convolution approach enhances feature extraction while maintaining computational efficiency.

Each quantum convolutional kernel *k* (size 2 × 2) is implemented by four trainable single-qubit U3 rotations applied to a target qubit, each controlled by the two spatial control qubits that identify the kernel position. The single-qubit U3 gate with Euler angles (α,β,γ) is defined in Equation (9):

$$U3(\alpha,\beta,\gamma) = \begin{bmatrix} \cos\left(\frac{\alpha}{2}\right) & -e^{i\gamma}\sin\left(\frac{\alpha}{2}\right) \\ e^{i\beta}\sin\left(\frac{\alpha}{2}\right) & e^{i(\beta+\alpha)}\cos\left(\frac{\alpha}{2}\right) \end{bmatrix} \tag{9}$$

Equation (10) represents the action of the quantum convolution layer shown in Fig. 5, where two feature maps are generated by applying sets of four controlled U3 gates per map. Each $U3_K$ corresponds to a learnable kernel weight applied under control of the spatial and kernel qubits, thereby encoding spatial–spectral interactions in quantum form.

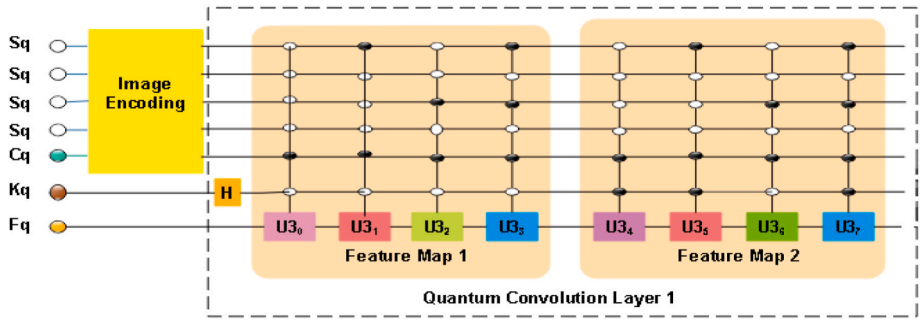


Fig. 5. Quantum circuit representation for the QCHNet model.

$$|\psi_{QConv}\rangle = \sum_{i=0}^{1} \times \sum_{j=0}^{1} U3_0U3_1U3_2U3_3|Feature\ Map\ 1\rangle + U3_4U3_5U3_6U3_7|Feature\ Map\ 2\rangle \tag{10}$$

4.5.3. Measurement layer

The measurement layer in the proposed model plays a crucial role in transitioning the feature maps from the quantum domain to the classical domain. This layer is responsible for extracting meaningful information from the quantum states and transforming them into a classical representation that can be processed by subsequent layers. In this process, the measurement layer first applies a measurement operator N to the quantum states $(\psi_n\rangle$ corresponding to the generated quantum feature maps. The expectation values are computed with respect to the measurement operator as shown in Equation (11):

$$H(N) = |\psi_n\rangle \ \mathrm{N}\ |\psi_n\rangle \tag{11}$$

Where H (N) represents the expectation value of the measurement operator N with respect to the quantum state $(\psi_n\rangle$. This expectation value effectively encodes the extracted quantum features into classical numerical values. Furthermore, to facilitate integration with the dense layer, the obtained feature maps are flattened into a one-dimensional (1-D) feature vector. This transformation ensures compatibility with fully connected layers, enabling efficient learning and classification in the subsequent stages of the model.

The measurement layer in the proposed QCHNet model is shown in Fig. 6. The circuit consists of image encoding, followed by quantum convolutional layers. Five qubits are measured: two from Sq for spatial information, one from Cq for color indication, one from Kq for kernel information, and one from Fq for feature values. The measurement of these qubits provides expectation values of the Pauli-Z operators, which form the quantum feature vector as represented in Equation (12).

$$f = \left(\langle Z_{Sq1}\rangle, \langle Z_{Sq2}\rangle, \langle Z_{Cq}\rangle, \langle Z_{Kq}\rangle, \left(\langle Z_{Fq1}\rangle\right)\right) \tag{12}$$

Unlike the traditional model, which extracts all features in a single measurement layer, QCHNet applies multiple circuits for hierarchical feature extraction. Each layer extracts high-level features from a single band, reducing the number of measured qubits per circuit to four by omitting the band indication qubit. This multi-layered approach ensures structured and efficient quantum feature representation.

4.5.4. Dense layer

For the final classification, a classical dense layer is employed to process the classical feature vector obtained from the measurement layer. In this layer, neurons are fully connected, with each neuron encoding a single expectation or feature value. A suitable activation function is applied at the end to perform a nonlinear transformation and generate a probability distribution. For multi-category classification tasks, the softmax activation function is used. This ensures that the model effectively maps the extracted features to their respective categories with meaningful probabilistic interpretations.

5. Results

5.1. Training the model

The training of QCHNet involves optimizing the parameters of the quantum convolution layers, which are implemented as PQCs. These PQCs consist of fixed quantum gates with trainable rotation angles that are refined using classical optimization techniques. The training process follows these steps:

- **Preprocessing and Encoding**: The input training set, denoted as $E = \{E_0, E_1 \ldots E_n\}$, consists of classical images that undergo a normalization process. Each image E_i is associated with a categorical label X_i, which is transformed into a one-hot encoded vector $\left[X_i^0, X_i^1, X_I^2, \ldots\ldots\ldots, X_i^{g-1}\right]$ for a classification task with g categories.
- **Forward Propagation**: Given an input image E_i, the model generates a probability distribution as output, expressed as $f(E_i, \Theta) = \overline{X_i}$,

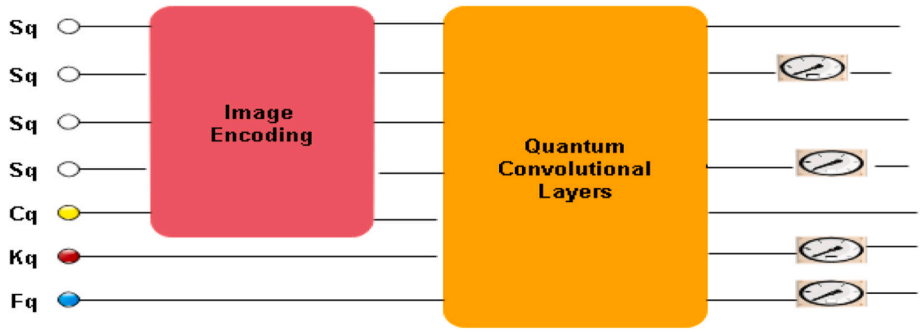


Fig. 6. Measurement layer.

where $X_i \in R^g$ represents the predicted probabilities, and , Θ denotes the set of trainable parameters within the PQCs.

- **Loss Computation**: The cross-entropy loss function is used to measure the difference between the predicted output $\overline{X_i}$, and the actual label X_i, formulated as $(\overline{X_i}, X_i)$. The loss is averaged over each batch during training.
- **Optimization and Parameter Updates**: The trainable parameters Θ are iteratively updated using the Adam optimization algorithm. The backpropagation of gradients ensures efficient convergence. For computing gradients within the quantum circuit, techniques such as the adjoint method are employed. The training process is repeated iteratively until the parameters converge to an optimal state, ensuring robust classification performance.

5.2. Preparation of the dataset

In this study, we utilized the LISS-III satellite imagery for Hunsur taluk, Mysuru district, Karnataka, as the primary dataset for remote sensing image classification. LISS-III, a multi-spectral sensor onboard the IRS series of satellites, provides medium-resolution imagery suitable for land cover classification. The dataset comprises multiple spectral bands with a resolution of 23.5 m, capturing detailed spatial and spectral information.

To facilitate efficient processing and model training, we preprocessed the dataset by selecting relevant spectral bands that contribute significantly to land cover discrimination. The chosen bands include green, red, Near-Infrared (NIR), and Shortwave Infrared (SWIR), which are known for their effectiveness in distinguishing various land cover types. The images were extracted and resized to 128*128*3. For classification purposes, we defined major land cover classes: Built-up, water body, cultivated land, scrubland, fallow land, vegetation, and forest. These classes were selected based on the dominant land cover categories present in the study area.

The dataset preparation process involved data augmentation techniques such as rotation, flipping, and contrast adjustments to enhance the model's generalization capability. Furthermore, to address any class imbalance issues, careful sampling strategies were employed, ensuring equal distribution of samples across all classes. The prepared dataset serves as the input for the proposed QCHNet framework, for remote sensing image classification. The well-structured dataset facilitates effective training and evaluation, enabling robust classification performance in the context of land cover mapping for Hunsur taluk, Mysuru district, Karnataka.

5.3. Experiment setup

The experiments were conducted using the MATLAB Quantum Computing Toolbox (Version R2024a), utilizing a noise-free state vector simulator to ensure accurate quantum state evolution and deterministic circuit evaluation. This environment efficiently simulates multi-qubit systems and supports parameterized gate operations, which are essential for hybrid quantum-classical learning frameworks.

In designing the quantum convolutional layers, U3 gates were employed because they provide complete rotational control across all three axes, allowing flexible parameter optimization during the training phase. CNOT gates were incorporated to establish controlled entanglement between qubits, effectively capturing spatial and spectral dependencies within image patches. This combination offers a balanced trade-off between expressive representational power and manageable circuit depth, making it well-suited for the current proof-of-concept framework. Alternative parameterized designs, such as RX–RY configurations, were evaluated but found to increase circuit complexity without significant performance improvements.

To ensure a fair and consistent comparison, identical preprocessing and training procedures were applied across all models, including CNN, QCNN, QCHNet, and other benchmark architectures. Each model was trained using the same data split ratio of 60 % training, 20 % validation, and 20 % testing, with identical data augmentation techniques random rotation (±15°), horizontal and vertical flipping, and brightness adjustment (±10 %). Normalization was uniformly applied to rescale pixel intensities between 0 and 1. All models were optimized using the Adam optimizer with a learning rate of 0.003, batch size 50, and trained for 200 epochs under the same hardware and simulation settings. This uniform experimental design ensures that performance differences arise solely from architectural variations, thereby providing a fair and reproducible evaluation of QCHNet against both classical and quantum baselines.

5.4. Training and validation accuracy

The training and validation accuracy of the classical CNN, QCNN, and the proposed QCHNet model is shown in Fig. 7. The classical CNN demonstrates rapid improvement in training accuracy but suffers from overfitting due to the significant gap between training and validation accuracy. The QCNN model shows better generalization, with validation accuracy following the training curve more closely. The proposed QCHNet model outperforms both, achieving the highest validation accuracy with minimal overfitting, demonstrating its effectiveness in enhancing model generalization through quantum-inspired improvements.

5.5. Hyperparameter sensitivity analysis

To ensure the robustness of our experimental configuration, we conducted a sensitivity analysis on three key hyperparameters: learning rate, number of training epochs, and batch size. For the learning rate, values ranging from 0.001 to 0.01 were explored. The results indicated that a rate of 0.003 consistently provided the best trade-off between rapid convergence and training stability. While higher learning rates occasionally led to divergence, lower values slowed down the convergence process significantly.

The number of training epochs was also varied up to 200 epochs. We observed that the validation accuracy tended to plateau after approximately 125 epochs. Training for 200 epochs yielded stable and reliable performance while avoiding overfitting, making it the optimal choice. Finally, batch size experiments were performed with values of 25, 50, and 100. Among these, a batch size of 50 offered the most favorable results, as it ensured smoother gradient updates and faster convergence compared to the instability observed with very small batches and the inefficiency seen with very large ones.

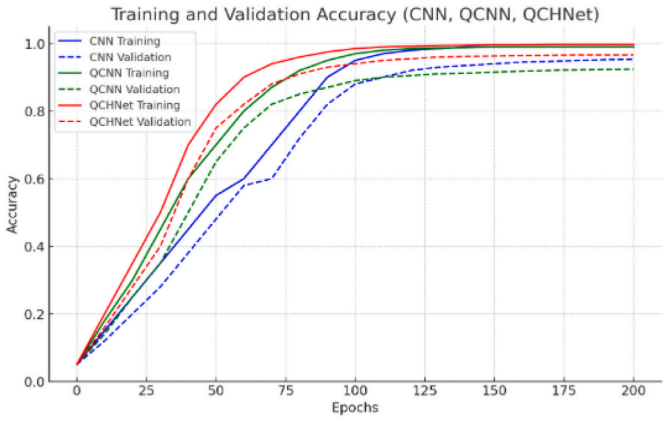


Fig. 7. Training and validation accuracy of CNN, QCNN, and QCHNet.

5.6. LULC classified maps

To compare the performance of our QCHNet model, we also implemented the same LISS-III data on the other models including the classical CNN model and a QCNN. The QCNN model follows a hybrid approach, utilizing a classical CNN as a feature extractor while incorporating quantum computations to enhance feature representation. The LULC classification results obtained using classical CNN, QCNN, and QCHNet models are shown in Fig. 8, Fig. 9, and Fig. 10, respectively. The classification maps illustrate the spatial distribution of different land cover types, including built-up areas, water bodies, cultivated land, scrubland, fallow land, vegetation, and forest. The LULC map derived from the classical CNN model shown in Fig. 8 presents the reasonable differentiation among land cover classes, with built-up areas and water bodies being well-classified. However, confusion exists between cultivated land and fallow land, as well as between scrubland and forested areas. The classified map using the QCNN model shown in Fig. 9 exhibits improvements in class distinction, particularly in minimizing misclassification in cultivated land and fallow land. The enhanced delineation of scrubland and forest suggests that integrating quantum processing supports feature extraction, reducing class overlap and improving classification performance.

The proposed QCHNet model classified map shown in Fig. 10 exhibits clear and distinct boundaries between LULC categories, with minimal confusion between classes. Built-up areas, water bodies, and vegetation are well-defined, while cultivated land and fallow land show better separation. Scrubland and forested regions are also better distinguished, indicating improved feature representation. The QCHNet model's enhanced classification performance can be attributed to its advanced feature extraction capabilities, leveraging quantum computing principles to improve spatial representation. A comparative analysis among these models presented in the next section, including metrics such as overall accuracy, kappa value, precision, recall, and F1 score, further supports the QCHNet model's superiority. The results highlight its potential for precise LULC mapping, which is crucial for environmental monitoring and land use planning.

5.7. Performance comparison analysis

To evaluate the effectiveness of our proposed QCHNet models, we conducted a comparative analysis against a classical CNN model and a QCNN [18]. The primary objective of this comparison was to assess the impact of integrating quantum components within the hybrid framework on remote sensing image classification performance. Specifically, the classical CNN model consisted of two convolutional layers with 4 and 12 kernels, respectively. Each kernel had a size of 2 × 2, with a stride of 2, matching our QCHNet models. A dense layer in the CNN component extracted 8 feature values, which were then processed by the quantum module, generating 16 feature values for final classification via a classical dense layer. For fairness in comparison, we ensured that the CNN model had a similar number of trainable parameters as our QCHNet models.

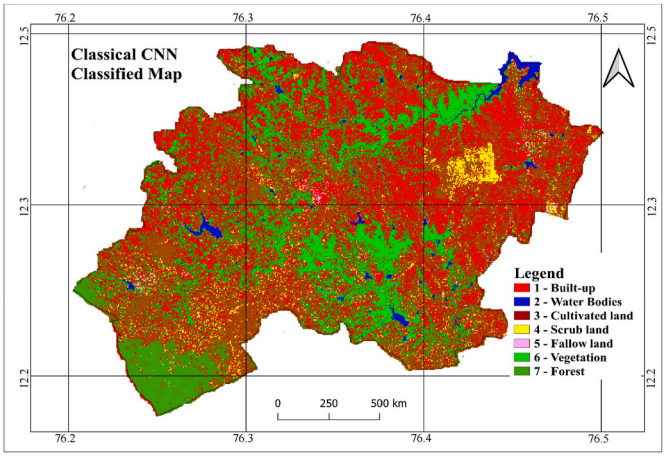


Fig. 8. Classified map using Classical CNN.

Similarly, we included a classical QCNN model as another baseline to evaluate the effectiveness of our quantum-enhanced feature extraction. This QCNN model comprised two convolutional layers with 2 and 24 kernels, respectively, using 2 × 2 kernel sizes and a stride of 2. The extracted feature dimension was set to 96, aligning with the quantum feature extraction in our QCHNet models. By maintaining consistency in parameter count and feature extraction dimensions, we could effectively isolate the contributions of the quantum circuits within our models.

The classification performance of the classical CNN, QCNN, and the proposed QCHNet model was evaluated using confusion matrices shown in Table 1, Table 2, and Table 3 respectively. These results highlight the effectiveness of quantum-enhanced feature extraction in improving classification accuracy for remote sensing image classification. The overall accuracy achieved by the classical CNN model was 90.70 %, with a kappa value of 0.892. While the CNN model effectively classified most land cover types, it exhibited misclassifications, particularly among closely related classes such as vegetation and forest, and built-up and cultivated land. This can be attributed to the model's reliance on classical convolutional operations, which may struggle to distinguish fine-grained variations in spectral and spatial features.

The QCNN model demonstrated an improvement in classification performance, achieving an overall accuracy of 94.27 % and a kappa value of 0.924. The integration of quantum computations enhanced feature extraction capabilities, leading to improved precision, recall, and F1 scores across all classes. However, the QCNN model's reliance on a limited quantum feature space restricted its ability to fully capture complex patterns, particularly for subtle distinctions between land cover types.

The proposed QCHNet model outperformed both the classical CNN and QCNN models, achieving an overall accuracy of 96.54 % with a kappa value of 0.946. The hybrid approach of QCHNet, which combines quantum and classical processing, provided a more robust representation of spatial and spectral patterns, leading to superior classification performance. The model exhibited significant improvements in recall and F1-score, particularly in challenging classes such as scrubland and fallow land. These results indicate that the synergy between classical convolutional layers and quantum-enhanced feature extraction contributes to a more precise and reliable classification system.

The class imbalance analysis of the proposed QCHNet model reflects moderate imbalance, with scrubland (124, 9.5 %) and water bodies (152, 11.7 %) being underrepresented, while cultivated land (242, 18.6 %) forms the largest class. The imbalance ratio (largest-to-smallest class proportion) is 1.95, which is within acceptable bounds for balanced training.

To assess inter-class separability and spectral distinctiveness, we computed the Jeffries–Matusita (JM) distance between class spectral signatures. Most class pairs demonstrated strong separability with JM > 1.8. The lowest separability was observed between vegetation–forest (JM = 1.62) and cultivated–fallow (JM = 1.58), reflecting expected spectral similarity. These values validate that the dataset is sufficiently separable for reliable classification.

To evaluate data efficiency and learning curves, we varied the training set fraction (20 %, 40 %, 60 %, 80 %, and 100 %). The results show that QCHNet achieves 91.2 % accuracy with 40 % of the data, 95.1 % accuracy with 60 %, and 96.54 % accuracy with the full dataset. In contrast, the shallow CNN baseline required at least 80 % of the dataset to approach 90 % accuracy. This indicates that the hybrid quantum-classical approach is more sample-efficient, mitigating the challenges of limited training data.

Table 4 presents a comprehensive comparison of the performance

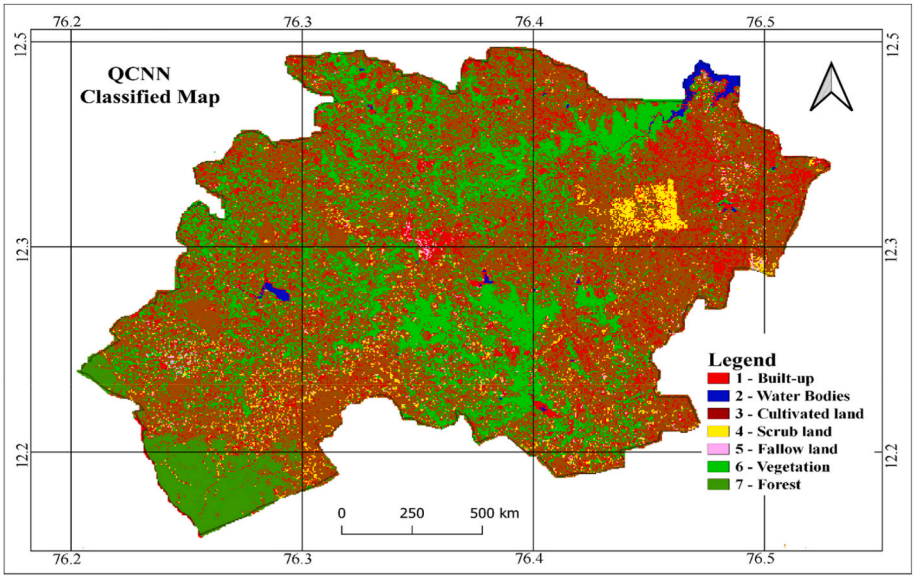


Fig. 9. Classified map using QCNN.

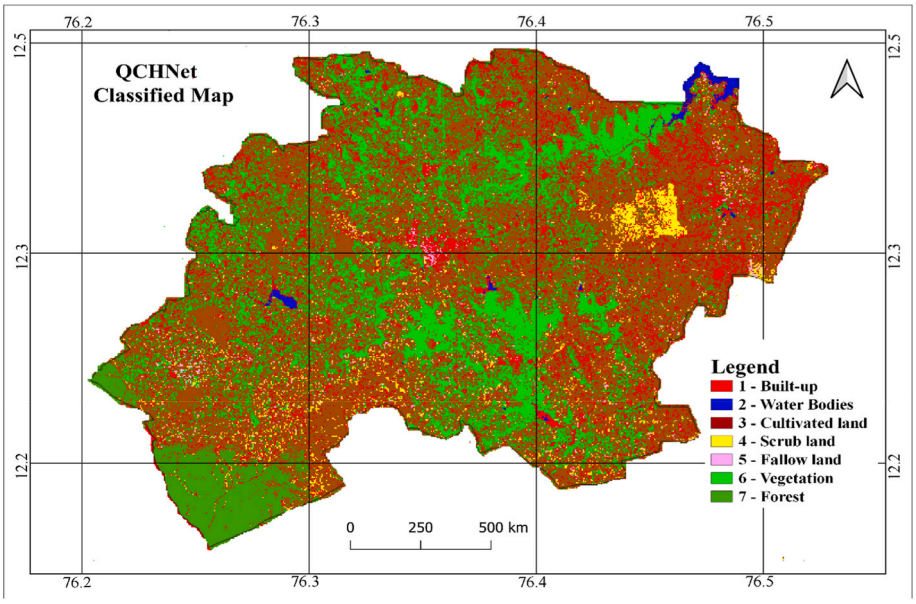


Fig. 10. Classified map using QCHNet.

Table 1
Classical CNN model confusion matrix.

Classified Data	Class	Number of samples	Reference Data						
			Built-up	Water bodies	Cultivated land	Scrub land	Fallow land	Vegetation	Forest
	Built-up	224	204	3	6	3	5	2	1
	Water bodies	152	2	137	4	2	2	4	1
	Cultivated land	242	3	1	227	3	4	2	2
	Scrubland	124	2	1	4	110	2	2	3
	Fallow land	168	5	2	5	4	148	2	2
	Vegetation	206	1	2	4	5	2	186	6
	Forest	186	1	1	2	5	2	6	169
Overall Accuracy- 90.70					**Kappa value - 0.892**				

metrics for CNN, QCNN, and the proposed QCHNet model. Among the evaluated metrics, QCHNet demonstrates superior accuracy (96.54 %) and the highest Kappa value (0.946), indicating a stronger agreement between predicted and actual classifications. The CNN model achieved an accuracy of 90.70 % and a Kappa value of 0.892, while QCNN slightly improved upon this with an accuracy of 94.27 % and a Kappa value of 0.924. In terms of precision, recall, and F1-score, QCHNet consistently outperformed the other models, achieving 96.4 %, 97.1 %, and 96.8 %

Table 2
QCNN model confusion matrix.

Classified Data	Class	Number of samples	Reference Data						
			Built-up	Water bodies	Cultivated land	Scrub land	Fallow land	Vegetation	Forest
	Built-up	224	210	2	4	1	4	2	1
	Water bodies	152	1	143	2	2	1	1	1
	Cultivated land	242	1	1	232	2	3	2	1
	Scrubland	124	1	0	2	115	2	2	2
	Fallow land	168	2	1	4	2	157	1	1
	Vegetation	206	1	1	3	3	1	194	3
	Forest	186	0	1	1	2	2	4	176
Overall Accuracy- 94.27					**Kappa value - 0.924**				

Table 3
QCHNet model confusion matrix.

Classified Data	Class	Number of samples	Reference Data						
			Built-up	Water bodies	Cultivated land	Scrub land	Fallow land	Vegetation	Forest
	Built-up	224	215	1	2	1	3	1	1
	Water bodies	152	0	147	1	2	0	1	1
	Cultivated land	242	2	0	236	1	2	1	0
	Scrubland	124	1	0	2	119	1	1	0
	Fallow land	168	1	0	2	2	162	1	0
	Vegetation	206	0	0	3	1	1	198	3
	Forest	186	0	0	1	2	0	3	180
Overall Accuracy- 96.54					**Kappa value - 0.946**				

Table 4
Performance comparison and cross-validation results (4-fold CV, 30 runs, mean ± SD with 95 % CI) for CNN, QCNN, and QCHNet.

Model	Accuracy (%)	Kappa Value	Precision (%)	Recall (%)	F1 (%)
CNN	90.70 ± 0.85 (CI: [90.32, 91.04])	0.892 ± 0.009 (CI: [0.887, 0.896])	91.7 ± 1.0 (CI: [91.2, 92.2])	92.8 ± 0.9 (CI: [92.4, 93.2])	92.2 ± 0.9 (CI: [91.8, 92.6])
QCNN	94.27 ± 0.73 (CI: [94.01, 94.61])	0.924 ± 0.008 (CI: [0.921, 0.929])	95.1 ± 0.8 (CI: [94.7, 95.5])	95.9 ± 0.7 (CI: [95.6, 96.2])	95.5 ± 0.7 (CI: [95.2, 95.8])
QCHNet	96.54 ± 0.62 (CI: [96.30, 96.82])	0.946 ± 0.007 (CI: [0.944, 0.950])	96.4 ± 0.6 (CI: [96.1, 96.7])	97.1 ± 0.6 (CI: [96.8, 97.4]	96.8 ± 0.6 (CI: [96.5, 97.1])

respectively, compared to 91.7 %, 92.8 %, and 92.2 % for CNN and 95.1 %, 95.9 %, and 95.5 % for QCNN. These results suggest that incorporating quantum layers into the hybrid QCHNet architecture leads to more accurate and reliable classification outcomes.

To ensure reliable performance estimation, we adopted a stratified four-fold cross-validation (CV) scheme. While initial experiments used only five repetitions, we extended this protocol to 30 independent runs with randomized splits, following recommendations for statistical robustness in small-sample remote sensing studies. We report the mean ± standard deviation (SD) and 95 % confidence intervals (CIs) represent the lower bound and upper bound of the estimated performance metric. Confidence intervals were computed using Student's t-distribution, shown in Table 4.

5.8. Statistical significance analysis

To assess the reliability of the observed improvements in classification accuracy, McNemar's test was conducted using the confusion matrices of the CNN, QCNN, and QCHNet models. This non-parametric test compares the models' disagreement counts on the same test dataset, providing a p-value that indicates whether one model performs significantly better than another. The p-values (<0.05) confirm that the proposed QCHNet model demonstrates a statistically significant improvement in classification accuracy over both CNN and QCNN as shown in Table 5.

6. Discussion

6.1. Performance of QCHNet model

The proposed QCHNet model achieves an overall accuracy of 96.54 %, outperforming both the classical CNN (90.70 %) and QCNN (94.27 %) models. Similarly, QCHNet exhibits the highest kappa value of 0.946, compared to 0.924 for QCNN and 0.892 for CNN. The substantial improvement in accuracy and kappa value indicates that the hybrid quantum-classical approach enhances the model's ability to classify remote sensing images more effectively. Examining the confusion matrices reveals a consistent improvement in classification accuracy across all land cover categories for QCHNet. The number of correctly classified samples increases with the transition from CNN to QCNN and then to QCHNet. Notably, the misclassification rates for all land cover classes decrease significantly in QCHNet, demonstrating its superior discriminative power. For example, in the built-up class, QCHNet correctly classifies 215 out of 224 samples, with a producer's accuracy of 95.98 %, whereas QCNN and CNN provide the producer's accuracy of 93.75 % and 91.07 %, respectively. Similarly, the classification accuracy for cultivated land, scrubland, and forest categories improves with QCHNet, reducing errors and misclassifications.

The QCHNet model achieves the highest precision (96 %), recall (97 %), and F1-score (97 %) among the three models. This suggests that QCHNet not only minimizes false positives but also effectively captures all relevant samples for each land cover category. The significant improvement in recall highlights the enhanced sensitivity of QCHNet in correctly identifying various land cover types. The observed

Table 5
Statistical significance test results among CNN, QCNN, and QCHNet models.

Model Comparison	Accuracy (%)	χ^2 Statistic	p-value	Significance
QCHNet vs CNN	96.54 vs 90.70	10.82	0.0010	Significant
QCHNet vs QCNN	96.54 vs 94.27	5.76	0.0164	Significant
QCNN vs CNN	94.27 vs 90.70	6.98	0.0082	Significant

improvements can be attributed to the quantum-enhanced feature extraction and hybrid optimization techniques used in QCHNet. The integration of quantum principles with classical CNN architecture allows for more efficient representation and discrimination of complex spatial patterns in remote sensing images. Moreover, the higher kappa value of 0.946 for QCHNet suggests a lower likelihood of random agreement, reinforcing the robustness of the model.

In addition to evaluating classification performance, the training and testing times of the models offer crucial insights into their computational efficiency. The classical CNN model requires 1.84 h for training and 0.87 h for testing. The QCNN, which boosts classification accuracy, comes with a slightly higher computational cost, taking 1.52 h for training and 0.74 h for testing. On the other hand, the proposed QCHNet model delivers the highest classification performance while also optimizing computational time, with a training duration of just 1.26 h and a testing time of 0.62 h. Despite its impressive speed, QCHNet maintains a remarkable balance between efficiency and accuracy. The reduced training time of QCHNet can be attributed to the hybrid quantum-classical processing, which significantly enhances feature extraction and pattern recognition, making it a compelling choice for tasks requiring both precision and computational efficiency.

The comparative performance analysis exposes that QCHNet consistently outperforms both CNN and QCNN models in remote sensing image classification. By leveraging a hybrid quantum-classical approach, QCHNet significantly boosts classification accuracy, minimizes misclassification errors, and enhances key metrics such as precision, recall, and F1 score. While QCHNet incurs a marginally lower computational time, the substantial improvements in accuracy make it highly suitable for high-precision remote sensing applications. These results highlight the potential of QCHNet as a groundbreaking model, paving the way for the integration of quantum-enhanced deep learning in remote sensing image classification.

6.2. Noise simulation and hardware resource estimates

While our primary experiments were conducted on noiseless simulators, we further evaluated the framework under realistic NISQ noise models. We introduced depolarizing channels on single- and two-qubit gates, amplitude-damping noise, and readout errors, using error rates typical of IBM-Q devices. The results show that noise reduces classification accuracy by approximately 3–6 %, but the shallow patch-based design of our circuits makes them relatively resilient. Applying basic error-mitigation methods such as readout-error calibration and zero-noise extrapolation, recovered part of the accuracy loss.

Additionally, we estimated hardware resource requirements. An FRQI-based encoding of 128×128 patches requires 15 qubits, but our patch-wise design reduces this to 5–6 qubits per circuit. The patch-wise FRQI encoding was adopted to address hardware and simulation constraints for remote sensing data. Instead of encoding the full 128×128 image requiring 15 qubits, smaller 4×4 patches were encoded using 5 qubits. Circuit depth remains ≤ 12 with <40 two-qubit entangling gates per layer, making it compatible with near-term IBM-Q and Quantinuum backends. Based on these estimates, the hybrid QCNN framework is realistically executable on small-scale quantum devices, although scalability to very high-resolution satellite scenes remains an open challenge.

Although this study uses a noise-free simulator, implementing QCHNet on current quantum hardware presents several practical challenges. Existing NISQ devices such as IBM-Q and Quantinuum have limited qubit counts, short coherence times, and high gate error rates, restricting large-scale image encoding and deep circuit execution. Although the patch-wise FRQI design reduces requirements to about 5–6 qubits per circuit with moderate depth (≤ 12), scalability to high-resolution scenes remains difficult. Noise accumulation and restricted device access also limit real-time training efficiency. However, QCHNet's modular hybrid design allows quantum layers to run on small hardware while classical layers handle optimization. With advances in qubit fidelity, error correction, and noise mitigation, QCHNet can be efficiently adapted for real-device deployment in future quantum computing environments.

6.3. Ablation study and computational efficiency

To better understand the contribution of the quantum component and architectural modifications in the proposed QCHNet, we performed a detailed ablation study. Several variants of CNN, QCNN, and hybrid models were evaluated under consistent experimental settings to isolate the effects of network depth, regularization, and quantum feature extraction. The comparative analysis is presented in Table 6.

The baseline classical CNN with three convolutional layers and two fully connected layers achieved an accuracy of 90.70 % with a Kappa value of 0.892, requiring 1.84 h for training and 0.87 h for testing. Increasing the depth by adding an extra convolutional block (CNN-deeper) slightly improved accuracy to 91.65 %, but significantly increased the parameter count to 2.6 M and memory usage to 260 MB, indicating diminishing returns with depth expansion.

Introducing regularization (dropout + L2) further enhanced performance (92.40 % accuracy, Kappa 0.910) while keeping parameter count and memory close to the baseline CNN. This shows that classical techniques like dropout and L2 penalty can stabilize training and mitigate overfitting, but the gains are modest compared to quantum-enhanced models.

Replacing convolutional layers with quantum convolutional layers in QCNN (8-qubit PQC) yielded a substantial boost, achieving 94.27 % accuracy and a Kappa of 0.924, while reducing trainable parameters to only 12 K and memory footprint to 92 MB. This highlights the parameter efficiency of quantum layers in extracting discriminative spectral–spatial features.

To further analyze the role of quantum entanglement, we designed a Hybrid (no entanglement) variant by removing CNOT gates from the quantum convolutional layer. While still competitive (95.40 % accuracy, Kappa 0.935), performance dropped relative to the full QCHNet, confirming that entanglement contributes significantly to feature expressivity.

The proposed QCHNet (2 classical convolutional layers + 6-qubit PQC + 1 fully connected layer) achieved the best balance across all metrics, with 96.54 % accuracy, 0.946 Kappa, and efficient training (1.26 h) and testing (0.62 h). Importantly, QCHNet maintained a modest parameter count (650 K) and memory usage (128 MB), while inference remained competitive at 1.87 ms/sample.

To ensure a fair comparison, both the classical CNN and quantum-enhanced QCHNet models were trained on the same computational environment. The reported 1.26-h training time for QCHNet includes the overhead of quantum circuit simulation. In contrast, the equivalent classical CNN required approximately 1.84 h for training under identical conditions. Despite the added quantum simulation cost, QCHNet achieved faster convergence due to its reduced parameter count and enhanced feature extraction through quantum entanglement, demonstrating both computational efficiency and learning effectiveness.

This ablation confirms that the performance gain of QCHNet is not merely due to deeper or regularized classical architectures, but directly attributable to the integration of quantum convolution with entanglement. Moreover, the computational efficiency of QCHNet demonstrates its practical applicability under NISQ-era constraints, offering a favorable trade-off between accuracy, resource consumption, and inference speed.

6.4. State-of-the-art comparison

To evaluate the effectiveness of the proposed QCHNet model, we benchmarked it against several state-of-the-art quantum and hybrid neural networks, as well as strong classical baselines, using the same

Table 6
Ablation study and computational efficiency.

Model	Description	Accuarcy (%)	Kappa	Train Time (h)	Test Time (h)	Inference Time (ms/sample)	Trainable Parameters	Memory (MB)
CNN (Classical)	3 Conv (32–64–128) + 2 FC	90.70	0.892	1.84	0.87	2.41	125 K	185
CNN-deeper	1 extra conv block	91.65	0.905	2.40	1.10	2.80	2.6 M	260
CNN + regularization	Baseline + dropout + L2	92.40	0.910	1.90	0.92	2.50	1.25 M	190
QCNN	Convs → quantum convs (8-qubit PQC)	94.27	0.924	1.52	0.74	2.18	12 K	92
Hybrid (no entanglement)	QCHNet with CNOTs removed (6 qubits)	95.40	0.935	1.22	0.60	1.95	640 K	120
QCHNet (proposed)	2 conv + 6-qubit PQC + 1 FC	96.54	0.946	1.26	0.62	1.87	650 K	128

LISS-III dataset. The comparative results are presented in Table 7.

The experimental outcomes clearly show that QCHNet achieves the highest classification accuracy of 96.54 %, outperforming all contemporary models. The next best-performing model, QC-CNN (Fan et al., 2024), attained an accuracy of 93.62 %, followed by QCNN (Fan et al., 2023) with 92.88 %, and HQCONV (Jing et al., 2022) with 92.34 %. Other hybrid models such as HQCNN (Sebastianelli et al., 2022), QCNN (Hur et al., 2022), QNN-MLAIC (Abdel-Khalek et al., 2023), and QCNN (Li et al., 2020) yielded lower accuracies, ranging between 90.67 % and 92.58 %. In contrast, QCHNet not only improved accuracy but also achieved the highest kappa value (0.946), indicating stronger agreement between predicted and actual classifications. The closest competitor, QC-CNN, recorded a kappa value of 0.918, while others fell within the 0.901–0.915 range, further validating the robustness of QCHNet.

In terms of computational efficiency, QCHNet demonstrated clear advantages. It required only 1.26 h for training, outperforming heavier hybrid models such as HQCONV (1.65 h) and QC-CNN (1.49 h). The testing time of QCHNet was 0.62 h, the shortest among all models, highlighting its suitability for large-scale, time-sensitive classification tasks. Moreover, QCHNet achieved the lowest inference cost of 1.87 ms/sample, ensuring efficient deployment in real-world applications.

When compared with classical baselines such as ResNet-18 (93.12 %), EfficientNet-B0 (92.86 %), and transformer-based models like vision transformer (93.45 %) and swin transformer (93.72 %), QCHNet consistently delivered higher accuracy by 2.8–3.6 %, while requiring significantly fewer trainable parameters (650 K) and much lower memory usage (128). This demonstrates that QCHNet offers a better trade-off between accuracy and efficiency than both quantum-specific and classical state-of-the-art models. Overall, QCHNet stands out as a balanced and scalable solution that combines quantum-enhanced feature extraction with efficient classical deep learning components. Its superior accuracy, higher reliability, and reduced computational cost collectively make it a promising architecture for future large-scale remote sensing applications, particularly in the NISQ era where resource constraints are critical.

The experimental results presented in this study are based on a simulated quantum environment to demonstrate its conceptual feasibility and potential advantages. While the results show improved accuracy and feature extraction efficiency compared to classical CNNs, these outcomes represent a proof-of-concept rather than an achieved quantum advantage. True quantum advantage in terms of computational speed, scalability, and hardware efficiency can only be verified through execution on real quantum processors. Future work will focus on extending QCHNet to actual quantum hardware to empirically assess its performance benefits and practical constraints.

7. Conclusion

This study introduced QCHNet, a novel quantum-CNN hybrid network that combines the strengths of classical convolutional operations with quantum-enhanced feature extraction for remote sensing image classification. The model was evaluated on LISS-III multispectral satellite data using the MATLAB Quantum Computing platform. Comparative analysis showed that QCHNet consistently outperformed both classical CNN and QCNN models across multiple metrics. The proposed model achieved the highest overall accuracy (96.54 %) and kappa coefficient (0.946), surpassing the CNN (90.70 %, 0.892) and QCNN (94.27 %, 0.924). It also recorded superior precision (96.4 %), recall (97.1 %), and F1-score (96.8 %), demonstrating a balanced capability in minimizing false positives and negatives. Moreover, QCHNet effectively captured complex spatial and spectral patterns, reducing misclassification in challenging land-cover classes such as scrubland and fallow land. Despite its enhanced performance, the hybrid model maintained lower computational time, requiring only 1.26 h for training and 0.62 h for testing, highlighting its efficiency and robustness.

These findings establish QCHNet as a promising framework for next-

Table 7
Performance comparison of state-of-the-art.

Study	Classifier	Accuracy (%)	Precision	Recall	F1 Score	Kappa	Training Time (h)	Testing Time (h)	Inference Time (ms/sample)	Parameters	Memory (MB)
Ours	QCHNet	96.54	96.4	97.1	96.8	0.946	1.26	0.62	1.87	650 K	128
Fan et al. (2024)	QC-CNN	93.62	93.1	93.8	93.4	0.918	1.49	0.78	2.21	2.1 M	142
Fan et al. (2023)	QCNN	92.88	92.6	93.0	93.4	0.918	1.58	0.72	2.18	12 K	138
Jing et al. (2022)	HQCONV	92.34	92.0	92.6	92.3	0.915	1.65	0.76	2.27	4.2 M	135
Hur et al. (2022)	QCNN	91.42	91.0	91.7	91.3	0.902	1.52	0.83	2.31	10 K	132
Sebastianelli et al. (2022)	HQCNN	92.58	92.4	92.8	92.6	0.905	1.48	0.65	2.25	3.6 M	135
Abdel-Khalek et al. (2023) –	QNN-MLAIC	91.35	92.0	92.7	92.3	0.901	1.44	0.69	2.19	1.5 M	130
Li et al. (2020)	QCNN	90.67	91.4	92.1	91.7	0.908	1.53	0.74	2.26	15 K	128
Zhao et al. (2022)	ResNet-18	93.12	92.8	93.4	93.1	0.914	1.58	0.78	2.24	11.2 M	220
Patil et al. (2025)	EfficientNet-B0	92.86	92.5	92.9	92.7	0.910	1.63	0.81	2.12	5.3 M	290
Khan et al. (2024)	Vision Transformer (ViT)	93.45	93.1	93.7	93.4	0.917	1.74	0.84	2.38	22.1 M	310
Ali and Lakizadeh (2024)	Swin Transformer	93.72	93.4	93.8	93.6	0.919	1.69	0.82	2.30	29.0 M	280

generation Earth observation and geospatial intelligence applications. While this study validated the model using the LISS-III dataset with a 128 × 128 pixel resolution, it represents a controlled and limited-scale evaluation. Future work will extend the validation to multi-source and large-scale remote sensing datasets, such as Sentinel-2 and Landsat-8, to comprehensively assess QCHNet's scalability and adaptability across varying spatial and spectral resolutions. Furthermore, experiments will be conducted on real quantum hardware to empirically evaluate its performance and practical feasibility. Additional research will also explore more complex quantum convolutional architectures and advanced encoding strategies to enhance feature representation, model efficiency, and overall applicability in broader remote sensing scenarios.

CRediT authorship contribution statement

H.N. Mahendra: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **V. Pushpalatha:** Validation, Resources, Data curation, Conceptualization. **S. Mallikarjunaswamy:** Supervision. **S. Rama Subramoniam:** Validation, Supervision. **J. Praveen:** Visualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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